

Practical no. 4 Modeling UML Use Case Diagrams and Capturing Use Case Scenarios

Aim:

The aim of this experiment is to model UML Use Case Diagrams and capture Use Case Scenarios for a Pharmacy Management System, helping to define the system's functionality and the interactions with external actors.

Introduction

A Use Case Diagram is a behavioral UML diagram that helps represent the functionality of a system, identifying the actors (users or external systems) and the use cases (functions or processes) they interact with. It provides a graphical overview that aids in understanding the system's requirements for both developers and stakeholders.

Objectives

By completing this exercise, you will:

- **Identify actors and use cases** from a given problem description.
- **Define relationships** (Include, Extend, and Generalization) between use cases.
- **Draw a Use Case Diagram** that represents the system's functionality and interactions with external entities.

Key Concepts:

1. **Actor:** An actor is any external entity (human, device, or system) that interacts with the system to achieve a goal. In a Pharmacy Management System, examples of actors include:
 - **Primary Actor:** Customer (who interacts directly with the system to buy medicines).
 - **Supporting Actor:** Pharmacist (manages inventory and processes orders).
 - **Admin:** Manages the entire system, including user management and database updates.
 - **Supplier:** Supplies medicines to the pharmacy.
2. **Use Case:** A use case represents a specific functionality or service the system provides. It can be an action that the system performs for an actor. For the Pharmacy Management System, use cases could include:

- **Register Customer** – A customer registers an account in the system.
- **Login** – A customer logs into their account.
- **Search Medicine** – Customers search for medicines in the system.
- **Place Order** – A customer places an order for medicine.
- **Process Payment** – The system processes the customer's payment.
- **Check Inventory** – Pharmacist checks the stock of medicines.
- **Update Inventory** – Admin updates inventory after receiving a new stock from suppliers.

3. Relationships Between Use Cases:

- **Include Relationship:** Used when a use case always includes another use case as part of its behavior. For example, in the **Place Order** use case, the **Login** use case may always be included before placing an order.
- **Extend Relationship:** Used when one use case may optionally extend another. For example, **Verify Prescription** could be an extension to **Place Order** if the customer orders a prescription-based medicine.
- **Generalization Relationship:** Used when a derived use case inherits behavior from a parent use case. For example, **Online Payment** and **Cash on Delivery** could be specialized forms of the **Process Payment** use case.

Example of Pharmacy Management System

Actors:

1. **Customer** – Uses the system to register, login, search, place orders, and make payments.
2. **Pharmacist** – Checks and manages inventory, processes orders.
3. **Admin** – Manages user accounts and system settings.
4. **Supplier** – Supplies medicines to the pharmacy.

Use Cases:

1. **Register Customer** – Allows a customer to create an account.
2. **Login** – The customer logs in with their credentials.
3. **Search Medicine** – The customer searches for available medicines.
4. **Place Order** – The customer places an order for medicine.

5. **Process Payment** – The system processes the customer's payment.
6. **Check Inventory** – The pharmacist checks the availability of medicines.
7. **Update Inventory** – The admin updates the stock levels after receiving medicines.
8. **Reorder Medicine** – The pharmacist reorders stock when medicine runs out.

Relationships:

- **Login** is included in **Place Order** (since a customer must log in before placing an order).
- **Verify Prescription** could extend **Place Order** if prescription medicines are involved.
- **Check Inventory** could be generalized to **Update Inventory** for inventory management tasks.

Procedure:

1. **Identify Actors:** Define external entities that interact with the system (e.g., Customer, Pharmacist, Admin).
2. **Identify Use Cases:** List all actions or services the system will provide (e.g., Register, Search, Place Order).
3. **Map Functional Requirements:** Relate each requirement to a specific use case (e.g., "User Login" maps to the **Login** use case).
4. **Define Relationships:** Identify how use cases relate to each other (e.g., **Login** is included in **Place Order**).
5. **Draw the Use Case Diagram:** Use a diagramming tool to create a visual representation of the use cases and actors and their relationships.

Conclusion:

The Use Case Diagram for the Pharmacy Management System will help visually organize and define how the system operates, the actors involved, and their relationships. This structured approach will ensure that all user requirements are captured and that the system can be developed to meet user expectations, improving communication among developers, end-users, and stakeholders.